

Felipe Ramos Roque

Senior Automation & AI Engineer | Business Systems Integration | Python, SQL, LLM APIs (Claude, GPT)

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Full overlap with US business hours - available as an international contractor

PROFESSIONAL SUMMARY

Senior automation and AI engineer with 5+ years owning internal process automation end to end - process discovery, integration design, development, deployment, and production support. Connects business systems through custom Python services and pipelines, including Google Workspace APIs, ERP and warehouse systems, CRMs, and third-party REST APIs, with SQL and Pandas behind the reporting. Integrates LLM APIs - Anthropic Claude, OpenAI, Azure OpenAI Service - into operational workflows, and co-built a multi-tenant AI agent platform now serving 22 client brands in production on Kubernetes. Delivered 20+ production automations for healthcare provider operations, removing over 500 hours of manual processing per month and handling tens of thousands of records monthly, working with patient and provider data under least-privilege access and masking. Postgraduate in Artificial Intelligence for Developers. Fluent in English and Portuguese.

TECHNICAL SKILLS

Languages & Core: Python, SQL, TypeScript, JavaScript, Java, FastAPI, Node.js, Pandas, Automated Testing (Pytest), CI/CD Pipelines, Git

Business Systems Integration: Google Workspace APIs (Sheets, Drive, Gmail, Calendar), Google Apps Script, Service-Account Authentication & OAuth Scopes, ERP Systems, Warehouse Management (WMS), CRM Platforms, Ticketing Systems, REST API Design & Integration, Webhooks, WhatsApp Business Platform, Meta Graph API

AI & LLM Integration: Anthropic Claude API, OpenAI API, Azure OpenAI Service, Prompt Engineering, Tool Use & Function Calling, AI Agent Development, LangGraph, LangChain, Retrieval-Augmented Generation (RAG), Hybrid Retrieval (BM25 + FAISS), AI Guardrails, Agent Tracing (Langfuse, LangSmith), Behavioral Testing

Process Automation: RPA (Robotic Process Automation), Business-Process Reengineering, Process Discovery & Documentation, Workflow Orchestration, n8n, Scheduled Jobs & Pipelines, Selenium, OpenCV, Web Scraping

Data & Reporting: SQL Query Development, PostgreSQL, MongoDB, Redis, ETL, Data Warehouse Integration, Operational Dashboards, Reporting Automation

Cloud & Infrastructure: Kubernetes, Docker, Multi-Tenant Architecture, Cloud Deployment, GitLab CI, Monitoring & Error Handling

Sensitive Data Practice: Healthcare Provider Operations, Patient and Provider Data Handling, Least-Privilege Access, Credential Isolation, Masking in Logs and Dashboards, Human Approval Gates, LGPD (Brazilian data protection law)

Languages Spoken: Portuguese (Native), English (Fluent)

PROFESSIONAL EXPERIENCE

AI Engineer

BasicX Solutions - São Paulo, SP, Brazil | October 2025 - Present

- Co-build and operate a multi-tenant conversational AI platform on WhatsApp, deployed as isolated Kubernetes services for 22 client brands across healthcare, veterinary, retail marketplace, real estate, fitness, and beauty verticals.
- Integrate LLM APIs (Anthropic Claude, OpenAI, Azure OpenAI Service) into production workflows through a hand-built LangGraph state machine - intent classification, lead extraction and persistence, retrieval, a tool-calling agent loop, and human-handoff detection - over checkpointed conversation state.
- Build the operational automation around the platform: an ERP-to-WMS order fulfillment integrator, a deterministic payroll-billing audit engine with a human approval gate, and migrations of n8n no-code workflows into tested Python services.
- Automate internal reporting into Google Workspace, so business teams read pipeline output in the tools they already work in rather than in a separate system.
- Enforce a supervisor and guardrail layer over model output - brand voice, output format, age gating, and crisis-trigger detection - backed by a behavioral test suite.
- Instrument production services with Langfuse and LangSmith tracing over OpenTelemetry, and hold deployed configuration to the repository through Pydantic-validated per-tenant YAML and a CI assertion job.

Founder & Principal Consultant

FEROQ - Worldwide | 2021 - Present

- Run an independent technology consultancy delivering process automation, AI-driven workflows, and full-stack applications to clients across healthcare, finance, and logistics.
- Act as the single technical owner of each engagement with minimal supervision: requirements discovery with stakeholders, solution architecture, development, cloud deployment, monitoring, and ongoing support.
- Integrate client business systems - Google Workspace APIs, ERPs, CRMs, and third-party REST APIs - into custom Python and Node.js pipelines with automated tests and CI/CD.
- Convert current-state operations into automation specifications, scope, and acceptance criteria, working directly with the business owners of each process.

Senior RPA Developer

Fiiibo - Fortaleza, CE, Brazil | April 2024 - September 2025

- Delivered 20+ production automations for healthcare provider operations, owning each end to end from process discovery and documentation through development, deployment, and support.
- Replaced manual, repetitive back-office workflows across multiple business fronts - removing over 500 hours of manual processing per month, automating tens of thousands of records monthly, and eliminating a class of manual data-entry error.
- Connected each automation to the systems the business already ran on - Google Workspace APIs, internal web and desktop applications, and vendor portals - through REST APIs where they existed and Python with Selenium, Pandas, and OpenCV where they did not.
- Handled patient and provider data throughout under least-privilege access, isolated credentials, and masking in logs and dashboards, in line with Brazil's LGPD.
- Built AI agents and n8n automations running in production for internal teams and client accounts, and produced operational dashboards giving stakeholders visibility into throughput and failures.

RPA Analyst Developer

Cadmus IT Solutions - São Paulo, SP, Brazil | March 2022 - September 2022

- Automated employee onboarding and offboarding for internal operations, provisioning and revoking system access across HR, IT asset management, and finance platforms, working directly with process owners.
- Delivered roughly six automations in seven months - three built new, three inherited unfinished or failing and rebuilt - and led the Python RPA squad's technical work, guiding one to two developers and up to three interns while introducing the team's first standards for project structure, code review, and documentation.

Python Developer

BPA Technologies - São Paulo, SP, Brazil | December 2020 - March 2022

- Delivered end-to-end process automation into client operations across insurance, healthcare, financial services, logistics, and consumer products at an RPA-as-a-service provider, owning four to five automations from discovery through development, testing, deployment, and rollout on an in-house Python RPA framework.
- Automated web, desktop, and file-based workflows with Selenium, Pandas, Openpyxl, and OpenCV, integrated systems via REST and Google Workspace APIs, built SQL reporting and dashboards for process status and failures, and handled production support and defect resolution with key users.

SELECTED PROJECTS

support-agent - Python, Anthropic Claude API, LangGraph, Pytest, GitHub Actions

- Open-source operational agent at github.com/feliperaro/support-agent, built to show production practice rather than a demo: a classify-retrieve-act loop on the Claude API with tool-level access control that refuses privileged actions before identity is verified, and output guardrails that block invented pricing.
- Paired with an evaluation harness that scores behavior structurally - tool calls, refusals, trajectory order, retrieval grounding - rather than response wording, running on a fresh clone with no API key and gated in CI against a score floor.

EDUCATION

Postgraduate Degree, Artificial Intelligence for Developers

FIAP (Postech) - São Paulo, Brazil | August 2024 - June 2025

Associate Degree, Information Technology

Tamwood Careers College - Toronto, Canada | July 2023 - December 2023

Bachelor's Degree, Systems Analysis and Development

FIAP - São Paulo, SP, Brazil | February 2020 - December 2021

Technical Degree, Information Technology

Senac - Taboão da Serra, SP, Brazil | February 2019 - July 2020